STUDY GUIDE

Female Reproductive Physiology

HPO Axis · Menstrual Cycle · Hormones · Pregnancy · Menopause

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PHS 220 — Physiology

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§1 OVERVIEW & ANATOMY

The female reproductive system comprises a set of internal and external organs whose integrated function facilitates the monthly preparation for fertilisation, the process of gestation (pregnancy), and the delivery of offspring. A woman's reproductive years are defined as the interval between **menarche** (first menstrual period) and **menopause** (cessation of menses for 12 consecutive months).

Structure	Key Anatomical Points	Primary Function
Ovaries (×2)	Almond-shaped; ~3.5 × 2 × 1 cm; in ovarian fossa (lateral pelvic wall); supplied by ovarian arteries (from aorta) and uterine arteries	Gametogenesis (oogenesis); hormone production (oestrogen, progesterone, androgens, inhibin)
Fallopian Tubes (×2)	~10 cm; 4 segments: infundibulum (fimbriae), ampulla (site of fertilisation), isthmus, interstitium; ciliated mucosa	Transport of ovum from ovary to uterus; site of fertilisation; motility via cilia and smooth muscle
Uterus	Pear-shaped; ~7.5 × 5 × 2.5 cm; fundus, body, isthmus, cervix; retroperitoneal; anteverted/anteflexed in most women	Implantation and development of embryo/fetus; menstruation
Cervix	Inferior uterine segment; external os opens into vagina; contains mucus glands; lined by columnar → squamous epithelium (transformation zone)	Controls access to uterine cavity; mucus plug (cervical barrier); dilatation in labour
Vagina	Fibromuscular tube ~9 cm; rugae allow distension; acidic pH (3.5–4) due to lactobacilli	Birth canal; receives penis during coitus; menstrual efflux
External genitalia (Vulva)	Mons pubis, labia majora and minora, clitoris, vestibule, Bartholin's glands, hymen	Sexual arousal; urinary and vaginal orifices

§2 THE HYPOTHALAMO-PITUITARY-OVARIAN (HPO) AXIS

The hypothalamus is the integrative centre of the reproductive axis. It receives neural inputs from numerous brain regions (cortex, limbic system, olfactory centres) and peripheral signals (hormonal, nutritional, metabolic), which it integrates to generate the precisely timed hormonal pulses that drive the menstrual cycle.

2.1 GnRH — The Pulse Generator

Gonadotropin-Releasing Hormone (GnRH) is a decapeptide synthesised in the arcuate nucleus of the hypothalamus. It is released into the hypothalamo-pituitary portal blood in discrete, regular pulses — this pulsatility is absolutely essential for reproductive function.

Parameter	Follicular Phase	Luteal Phase	Clinical Note
GnRH pulse frequency	~1 pulse/70–90 min (higher frequency)	~1 pulse/3–4 hours (lower frequency)	High frequency → ↑ LH; Low frequency → ↑ FSH
Effect of continuous GnRH	Paradoxical inhibition	Paradoxical inhibition	Basis of GnRH agonist therapy (prostate cancer, endometriosis, assisted reproduction)
Pulse amplitude	Increases pre-ovulation	Higher amplitude, lower frequency	Progesterone (luteal) slows pulse frequency; oestrogen increases it
Modulators (stimulatory)	Norepinephrine, glutamate, kisspeptin (KNDy neurons)	—	Kisspeptin-neurokinin B-dynorphin (KNDy) neurons in arcuate nucleus are the master pulse generator
Modulators (inhibitory)	Dopamine, endogenous opioids (β-endorphin), cortisol (stress)	↑ Opioid tone in luteal phase	Stress → ↑ opioids → ↓ GnRH → menstrual irregularity

2.2 LH and FSH — Gonadotropins

Gonadotropin	Source	Primary Targets	Key Actions in Female
FSH (Follicle-Stimulating Hormone)	Pituitary gonadotrophs (shared cells with LH)	Granulosa cells of follicle	1) Stimulates follicle recruitment and growth; 2) Induces aromatase in granulosa cells (converts androgens → oestradiol — the "two-cell, two-gonadotropin" model); 3) Induces LH receptors on granulosa cells pre-ovulation; 4) Promotes inhibin B secretion
LH (Luteinising Hormone)	Pituitary gonadotrophs	Theca cells; granulosa cells; corpus luteum	1) Stimulates theca cells to produce androgens (androstenedione) as substrate for aromatase; 2) LH surge triggers ovulation (~36 hours later); 3) Luteinises granulosa cells → corpus luteum formation; 4) Stimulates corpus luteum to produce progesterone

THE TWO-CELL, TWO-GONADOTROPIN MODEL

Oestradiol synthesis in the follicle requires cooperation between two cell types and both gonadotropins:
1. LH acts on theca cells → cholesterol → androgens (androstenedione, testosterone) via CYP17A1.
2. Androgens diffuse to adjacent granulosa cells, where FSH upregulates aromatase (CYP19) → androgens → oestradiol.
Neither cell type can make oestradiol alone — cooperation is obligatory. This explains why FSH and LH deficiency each impair oestrogen production.

2.3 Feedback Mechanisms

Hormone	Effect on Pituitary/Hypothalamus	Phase
Oestradiol (low-moderate levels)	Negative feedback: ↓ LH and FSH (via ↓ GnRH sensitivity and direct pituitary suppression)	Follicular phase (most of cycle)
Oestradiol (high sustained levels, >200 pg/mL for >36h)	Positive feedback: ↑↑↑ LH surge — paradoxical switch from inhibition to stimulation	Late follicular phase → OVULATION
Progesterone (luteal phase)	Negative feedback: ↓ GnRH pulse frequency; combined with oestrogen strongly suppresses LH/FSH	Luteal phase
Inhibin B (from granulosa cells)	Selectively suppresses FSH (not LH)	Follicular phase
Inhibin A (from corpus luteum)	Suppresses FSH; reinforces negative feedback in luteal phase	Luteal phase

§3 REPRODUCTIVE HORMONES

3.1 Oestrogens

Type	Potency	Main Source	Phase
Oestradiol (E2)	Most potent (1×)	Granulosa cells (follicle/corpus luteum); placenta	Reproductive years (dominant)
Oestrone (E1)	Less potent (~30% of E2)	Peripheral conversion of androstenedione in adipose tissue	Post-menopause (dominant oestrogen)
Oestriol (E3)	Weakest (10% of E2)	Placenta (from fetal DHEAS)	Pregnancy (predominant)

System	Oestradiol Effects
Reproductive tract	↑ Endometrial proliferation; ↑ vaginal rugae and lubrication; cervical mucus becomes thin and watery (sperm-penetrable) at ovulation; breast development (ductal proliferation)
Bone	↑ Bone density (inhibits osteoclasts; ↑ OPG; ↑ calcitonin); maintains epiphyseal closure; loss at menopause → osteoporosis
Cardiovascular	Cardioprotective premenopausally: ↑ HDL; ↓ LDL; ↑ NO (vasodilation); ↓ atherosclerosis risk vs males
Metabolism	↑ LBM (lean body mass); reduces visceral adiposity; ↑ insulin sensitivity; ↑ clotting factors (II, VII, IX, X — risk of VTE with exogenous oestrogen)
CNS	Neuroprotective; modulates mood (↑ serotonin); affects thermoregulation (loss at menopause → hot flushes)
Liver	↑ SHBG; ↑ TBG; ↑ coagulation factors; ↑ angiotensinogen (→ ↑ BP with OCP)

3.2 Progesterone

Progesterone is a C-21 steroid produced primarily by the corpus luteum after ovulation (luteal phase levels: 15–30 nmol/L). It is the hormone of the luteal phase and pregnancy. Its actions are largely complementary to oestrogen — the two hormones work sequentially to prepare the uterus for implantation.

Target	Progesterone Effect
Endometrium	Transforms proliferative endometrium (oestrogen-primed) → secretory, decidualised endometrium; glands become coiled and secretory; essential for implantation
Myometrium	↓ Uterine contractility (quiescence essential for maintaining pregnancy)
Cervix	↓ Cervical mucus becomes thick, viscous, hostile to sperm (post-ovulation); closes cervical os
Breast	Promotes lobular-alveolar development (milk glands); with oestrogen → full breast development
Hypothalamus	↑ Basal body temperature by 0.2–0.5°C (used to detect ovulation); ↓ GnRH pulse frequency
Breathing	↑ Respiratory drive (↑ sensitivity to CO_2) — relevant in pregnancy: physiological hyperventilation

DAY 21 PROGESTERONE TEST

A serum progesterone measured 7 days before the next expected period (conventionally Day 21 in a 28-day cycle) is used to confirm ovulation. A level >30 nmol/L (16 ng/mL) confirms ovulation occurred. In irregular cycles, sampling is done weekly until menstruation occurs. Reference ranges: Follicular phase: <3 nmol/L; Post-ovulatory (luteal): 15–160 nmol/L; Post-menopausal: <3 nmol/L.

3.3 Prolactin

Prolactin is a 199-amino acid polypeptide secreted by lactotroph cells of the anterior pituitary. Uniquely, its secretion is under tonic inhibitory control by dopamine (prolactin-inhibiting hormone, PIH) from the hypothalamus.

Aspect	Details
Stimulators of prolactin	Suckling (via neurogenic reflex → ↓ dopamine → ↑ prolactin); TRH; VIP; oestrogen; stress; sleep; antipsychotics (dopamine antagonists); metoclopramide; methyldopa
Inhibitors of prolactin	Dopamine (dominant, tonic); somatostatin; bromocriptine/cabergoline (dopamine agonists — treat hyperprolactinaemia)
Actions of prolactin	Milk synthesis (alveolar cells); ↑ growth of mammary glands; suppresses GnRH → anovulation → lactational amenorrhoea (physiological contraception); immunomodulatory

Aspect	Details
Hyperprolactinaemia	Causes: prolactinoma (most common); drugs; hypothyroidism (↑ TRH stimulates prolactin); renal failure. Features: oligo/amenorrhoea; galactorrhoea; infertility; reduced libido
Measurement	Indicated in: oligomenorrhoea, amenorrhoea, galactorrhoea, subfertility, hypogonadotrophic hypogonadism

§4 THE MENSTRUAL CYCLE

The menstrual cycle averages 28 days (range 21–35 days) and is divided into three functional phases based on events in the uterus (follicular, ovulation, luteal) or equivalently into the follicular (proliferative) and luteal (secretory) phases relative to ovulation. Day 1 of the cycle is defined as the first day of menstrual bleeding.

Phase	Days (28-day cycle)	Dominant Hormone	Ovarian Events	Endometrial Events
Menstruation	Days 1–5	↓ E2 and Progesterone	Corpus luteum degenerates (luteolysis); new follicle recruitment begins	Functional layer shed; prostaglandin-mediated ischaemia and spiral arteriole spasm → shedding
Follicular (Proliferative)	Days 6–13	Rising Oestradiol (FSH-driven)	Primary follicle → secondary → Graafian follicle; dominant follicle selected (~10mm) by Day 7; suppresses others via ↑ E2 and inhibin B	Endometrium proliferates; glands elongate; spiral arteries develop; thickness ↑ to 8–12mm
Ovulation	Day 14 (±2)	LH surge (triggered by high E2)	LH surge → oocyte maturation; follicle rupture → release of secondary oocyte; LH surge ~36h before ovulation	Cervical mucus becomes thin and elastic (spinnbarkeit); ferning pattern on microscopy
Luteal (Secretory)	Days 15–28	Progesterone (dominant) + E2	Corpus luteum forms from ruptured follicle granulosa cells (luteinised by LH); peaks Days 19–23; degenerates Day 24–26 if no implantation	Endometrium becomes secretory; glands coiled; stromal oedema; spiral arteries coil — optimal for implantation

THE LH SURGE — MECHANISM

As the dominant follicle grows, oestradiol rises progressively. When E2 remains >200 pg/mL for ≥36–48 hours, a critical switch occurs in the pituitary: instead of negative feedback, oestrogen triggers massive positive feedback → GnRH receptors upregulate → an explosive LH surge (peak ~100 mIU/mL). The LH surge:
1. Resumes meiosis I in the oocyte (from dictyotene arrest).
2. Stimulates prostaglandin synthesis in the follicle wall → chemotaxis of proteases (collagenase) → follicle wall digestion → rupture.
3. Luteinises granulosa cells → corpus luteum formation.

§5 OOGENESIS & FOLLICULOGENESIS

Stage	Cell	Timing	Key Event
Oogonia	Mitotically active diploid cells	Foetal life (3–7 weeks); peak at 20 weeks (6–7 million)	Maximal germ cell number; then massive atresia
Primary Oocyte	Enters meiosis I; arrested in prophase I (diplotene/dictyotene)	Foetal → persists for decades until ovulation	Arrest maintained by OMI (oocyte maturation inhibitor from granulosa cells)
Primordial Follicle	Primary oocyte + single layer of flattened granulosa cells	At birth: ~1–2 million; puberty: ~400,000; menopause: ~1,000	Non-growing pool; recruited from birth onwards (atresia > growth); only 400–500 ovulate in lifetime
Primary Follicle	Oocyte + cuboidal granulosa cells; zona pellucida forms	Months–years after recruitment	FSH-independent initial recruitment; zona pellucida is acellular glycoprotein coat (ZP1-3)
Secondary Follicle	+ Theca cells (inner and outer) develop; multiple granulosa layers; antrum begins	FSH-responsive stage	Antrum (fluid-filled cavity) appears; granulosa cells make oestradiol; theca cells make androgens
Graafian (Tertiary) Follicle	Large antrum; cumulus oophorus; oocyte surrounded by corona radiata	Final days before ovulation; FSH-sensitive	Dominant follicle selected by increased FSH sensitivity; all others undergo atresia
Ovulation	Secondary oocyte + first polar body	Day 14 (LH surge → ovulation ~36h later)	Meiosis I completed; arrested at metaphase II until fertilisation (if it occurs)
Corpus Luteum	Luteinised granulosa + theca cells	Post-ovulation; lifespan 14 days unless rescued by hCG	Secretes progesterone (+E2); rescued by hCG in pregnancy → persists ~10 weeks
Corpus Albicans	Fibrotic scar tissue	If no fertilisation: luteolysis Day 24–26	Progesterone falls → endometrium shed → menstruation begins

§6 PREGNANCY & IMPLANTATION

Event	Timing	Details
Fertilisation	Within 12–24h of ovulation	Ampulla of fallopian tube; acrosome reaction; zona pellucida penetration; cortical reaction (block to polyspermy)
Cleavage / Morula	Days 1–4 post-fertilisation	Mitotic divisions without growth; 16-cell morula by Day 4
Blastocyst formation	Day 5	Inner cell mass (ICM → embryo) + outer trophoblast layer
Implantation	Days 6–10	Blastocyst hatches from zona; trophoblast invades decidualised endometrium (secretory phase); syncytiotrophoblast invasive
hCG secretion begins	Day 8–10 (post-fertilisation)	Syncytiotrophoblast secretes hCG → rescues corpus luteum → sustained progesterone → maintains pregnancy
hCG peak	Weeks 8–10 of pregnancy	Basis of pregnancy tests; hCG stimulates thyroid (structural similarity to TSH → gestational hyperthyroidism)

Hormone	Source	Key Roles in Pregnancy
hCG	Syncytiotrophoblast	Rescues corpus luteum; stimulates fetal testosterone (male sex differentiation); TSH-like → mild thyroid stimulation
hPL (Placental Lactogen)	Placenta	Insulin antagonist → ↑ maternal blood glucose for fetus; ↑ lipolysis; mammotropic (breast preparation); GH-like growth effects
Progesterone	Corpus luteum → Placenta (from week 8–10 — "luteo-placental shift")	Myometrial quiescence; immunotolerance of fetus; cervical plug; prevents premature labour
Oestrogen (oestriol)	Placenta (from fetal adrenal DHEAS → placental aromatase)	Uterine growth; ↑ breast growth; ↑ oxytocin receptors (term); ↑ uteroplacental blood flow
Oxytocin	Posterior pituitary (maternal)	Uterine contractions (Ferguson reflex); positive feedback loop → parturition; milk let-down reflex

§7 MENOPAUSE

Menopause is defined as the permanent cessation of menstruation due to loss of ovarian follicular activity, confirmed after 12 consecutive months of amenorrhoea. The average age in the UK is 51 years (range 45–55). The perimenopause (climacteric) begins 2–8 years before menopause and is characterised by irregular cycles as the ovarian reserve depletes.

Parameter	Change at Menopause	Clinical Consequence
Ovarian follicles	Depleted → no oestradiol production	Central change — all other consequences follow from this

Parameter	Change at Menopause	Clinical Consequence
Oestradiol (E2)	↓↓↓ (drops 90%; <40 pmol/L post-menopause vs ~400 follicular)	Hot flushes; vaginal atrophy; osteoporosis; cardiovascular risk ↑
FSH	↑↑ (>25 IU/L; often >100 IU/L in established menopause)	Used diagnostically; reflects loss of inhibin B and oestradiol negative feedback
LH	↑ (less dramatic than FSH)	Both FSH and LH become very high ("hypergonadotropic hypogonadism")
Inhibin B	↓ (early marker of diminishing ovarian reserve)	Falls before E2 → FSH rises — first laboratory sign of perimenopausal transition
Androgen levels	Mild ↓ (ovarian stromal still some T, but adrenal contribution persists)	Post-menopausal women still have androgens → oestrone from peripheral aromatisation of androstenedione

SYMPTOMS OF MENOPAUSE

Vasomotor: Hot flushes (80%); night sweats; palpitations. Caused by ↓ E2 → disinhibition of hypothalamic thermoregulatory centre → inappropriate heat dissipation response.
Genitourinary: Vaginal dryness; dyspareunia; atrophic vaginitis; urinary frequency/urgency (GSM — genitourinary syndrome of menopause).
Psychological: Mood changes; depression; anxiety; cognitive difficulties; insomnia.
Skeletal: Accelerated bone loss (up to 3–5% per year in first 5 years) → osteoporosis → fractures (vertebral, hip, Colles).
Cardiovascular: Loss of cardioprotection → ↑ LDL; ↑ atherosclerosis; ↑ CVD risk post-menopausally.

§8 CLINICAL CORRELATES

Condition	Definition/Aetiology	Key Features	Investigation Pattern
PCOS (Polycystic Ovary Syndrome)	Commonest endocrine disorder in women (10%); Rotterdam criteria (2 of 3): oligomenorrhoea/anovulation; hyperandrogenism (clinical or biochemical); polycystic ovaries on USS	Oligomenorrhoea; hirsutism; acne; obesity; infertility; insulin resistance (risk of T2DM); LH:FSH ratio >2:1 (classically)	Elevated LH; Normal/low FSH; ↑ free testosterone; ↑ DHEAS; ↑ LH:FSH; USS 12+ follicles per ovary
Premature Ovarian Insufficiency (POI)	Loss of ovarian function before age 40 (previously called "premature menopause")	Oligomenorrhoea/amenorrhoea; hot flushes; infertility; osteoporosis risk; causes: autoimmune, iatrogenic (chemo/radio), chromosomal (Turner's XO)	FSH >25 IU/L on two occasions 4 weeks apart; low E2
Hypothalamic Amenorrhoea	Functional GnRH deficiency due to calorie restriction; excessive exercise; stress; weight loss	Amenorrhoea; low BMI; low E2; no LH surge	Low FSH; Low LH; Low E2; Low-normal prolactin

Condition	Definition/Aetiology	Key Features	Investigation Pattern
Hyperprolactinaemia	Prolactinoma; drugs; hypothyroidism; idiopathic	Galactorrhoea; oligomenorrhoea; infertility; reduced libido; visual field defects (macroadenoma)	Elevated prolactin; low LH and FSH (prolactin inhibits GnRH); MRI pituitary